1.Consider the Insurance database given below.
The primary keys are underlined and the data types are specified:
PERSON (driver – id #: String, name: string, address: string)
CAR (regno: string, model: string, year: int)
ACCIDENT (report-number: int, accd-date: date, location: string)
OWNS (driver-id #:string, Regno:string)
PARTICIPATED (driver-id: string, Regno:string, report-number:int, damage amount:int)
(i). Create the above tables by properly specifying the primary keys and the foreign keys.
(ii) Enter at least five tuples for each relation.
(iii) Demonstrate how you
a. Update the damage amount to 25000 for the car with a specific Regno in the ACCIDENT table with report number 12.
b. Add a new accident to the database.
(iv) Find the total number of people who owned cars that were involved in accidents in 2008.
(v) Find the number of accidents in which cars belonging to a specific model were involved.
(vi) Generate suitable reports.
(vii) Create suitable front end for querying and displaying the results.

```
mysql> CREATE DATABASE QUESTION1;
Query OK, 1 row affected (0.01 sec)

mysql> USE QUESTION1;
Database changed
mysql> SHOW TABLES;
Empty set (0.01 sec)
```

```
mysql> CREATE TABLE PERSON (
    ->     DRIVER_ID VARCHAR(10) PRIMARY KEY,
    ->     NAME VARCHAR(20) NOT NULL,
    ->     ADDRESS VARCHAR(30)
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql>
mysql> CREATE TABLE CAR (
    ->     REG_NO VARCHAR(10) PRIMARY KEY,
    ->     MODEL VARCHAR(20) NOT NULL,
    ->     YEAR INT
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> CREATE TABLE ACCIDENT (
    ->     REPORT_NO INT PRIMARY KEY,
    ->     DDATE DATE,
    ->     LOCATION VARCHAR(30) NOT NULL
    -> );
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> CREATE TABLE OWNS (
    ->     DRIVER_ID VARCHAR(10),
    ->     REG_NO VARCHAR(10),
    ->     PRIMARY KEY (DRIVER_ID, REG_NO),
    ->     FOREIGN KEY (DRIVER_ID) REFERENCES PERSON(DRIVER_ID) ON DELETE CASCADE,
    ->     FOREIGN KEY (REG_NO) REFERENCES CAR(REG_NO) ON DELETE CASCADE
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> CREATE TABLE PARTICIPATED (
    ->     DRIVER_ID VARCHAR(10),
    ->     REG_NO VARCHAR(10),
    ->     REPORT_NO INT,
    ->     DAMAGE_AMOUNT INT,
    ->     PRIMARY KEY (DRIVER_ID, REG_NO, REPORT_NO),
    ->     FOREIGN KEY (DRIVER_ID) REFERENCES PERSON(DRIVER_ID) ON DELETE CASCADE,
    ->     FOREIGN KEY (REG_NO) REFERENCES CAR(REG_NO) ON DELETE CASCADE,
    ->     FOREIGN KEY (REPORT_NO) REFERENCES ACCIDENT(REPORT_NO) ON DELETE CASCADE
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> SHOW TABLES;
+---------------------+
| Tables_in_question1 |
+---------------------+
| accident            |
| car                 |
| owns                |
| participated        |
| person              |
+---------------------+
5 rows in set (0.00 sec)
```

```
mysql> -- PERSON
mysql> INSERT INTO PERSON VALUES ('D001', 'ADARSH', 'JPNAGAR');
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO PERSON VALUES ('D002', 'ANOOP', 'JAYANAGAR');
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO PERSON VALUES ('D003', 'KIRAN', 'JAYANAGAR');
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO PERSON VALUES ('D004', 'ABHISHEK', 'PEENYA');
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO PERSON VALUES ('D005', 'SANJAY', 'RAJAJINAGAR');
Query OK, 1 row affected (0.00 sec)
```

```
mysql> SELECT * FROM PERSON;
+------------+----------+-------------+
| DRIVER_ID  | NAME     | ADDRESS     |
+------------+----------+-------------+
| D001       | ADARSH   | JPNAGAR     |
| D002       | ANOOP    | JAYANAGAR   |
| D003       | KIRAN    | JAYANAGAR   |
| D004       | ABHISHEK | PEENYA      |
| D005       | SANJAY   | RAJAJINAGAR |
+------------+----------+-------------+
5 rows in set (0.00 sec)
```

```
mysql>
mysql> -- CAR
mysql> INSERT INTO CAR VALUES ('KA01M1001', 'SKODA', 2002),
    ->                                       ('KA02B2002', 'BMW', 2008),
    ->                          ('KA03M3003', 'MARUTI', 2008),
    ->                          ('KA04H4004', 'HYUNDAI', 2001),
    ->                          ('KA05A5005', 'AUDI', 2008);
Query OK, 5 rows affected (0.02 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql>
mysql> -- ACCIDENT
mysql> INSERT INTO ACCIDENT VALUES (1, '2002-01-01', 'RJNGR'),
    ->                                                (2, '2003-02-03', 'JNGR'),
    ->                                                (3, '2003-04-05', 'PEENYA'),
    ->                                                (12, '2008-05-15', 'MGROAD'),
    ->                                                (15, '2008-08-20', 'JPNGR');
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

```
mysql>
mysql> -- OWNS
mysql> INSERT INTO OWNS VALUES ('D001', 'KA01M1001'),
    ->                                        ('D002', 'KA02B2002'),
    ->                                        ('D003', 'KA03M3003'),
    ->                                        ('D004', 'KA04H4004'),
    ->                                        ('D005', 'KA05A5005');
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql>
mysql> -- PARTICIPATED
mysql> INSERT INTO PARTICIPATED VALUES ('D001', 'KA01M1001', 1, 9000),
    ->                                                    ('D002', 'KA02B2002', 12, 15000),
    ->                                 ('D003', 'KA03M3003', 12, 10000),
    ->                                 ('D004', 'KA04H4004', 3, 67800),
    ->                                 ('D005', 'KA05A5005', 15, 20000);
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM PERSON;
+-----------+----------+-------------+
| DRIVER_ID | NAME     | ADDRESS     |
+-----------+----------+-------------+
| D001      | ADARSH   | JPNAGAR     |
| D002      | ANOOP    | JAYANAGAR   |
| D003      | KIRAN    | JAYANAGAR   |
| D004      | ABHISHEK | PEENYA      |
| D005      | SANJAY   | RAJAJINAGAR |
+-----------+----------+-------------+
5 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM CAR;
+-----------+---------+------+
| REG_NO    | MODEL   | YEAR |
+-----------+---------+------+
| KA01M1001 | SKODA   | 2002 |
| KA02B2002 | BMW     | 2008 |
| KA03M3003 | MARUTI  | 2008 |
| KA04H4004 | HYUNDAI | 2001 |
| KA05A5005 | AUDI    | 2008 |
+-----------+---------+------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM ACCIDENT;
+-----------+------------+----------+
| REPORT_NO | DDATE      | LOCATION |
+-----------+------------+----------+
|         1 | 2002-01-01 | RJNGR    |
|         2 | 2003-02-03 | JNGR     |
|         3 | 2003-04-05 | PEENYA   |
|        12 | 2008-05-15 | MGROAD   |
|        15 | 2008-08-20 | JPNGR    |
+-----------+------------+----------+
5 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM OWNS;
+-----------+-----------+
| DRIVER_ID | REG_NO    |
+-----------+-----------+
| D001      | KA01M1001 |
| D002      | KA02B2002 |
| D003      | KA03M3003 |
| D004      | KA04H4004 |
| D005      | KA05A5005 |
+-----------+-----------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM PARTICIPATED;
+-----------+-----------+-----------+---------------+
| DRIVER_ID | REG_NO    | REPORT_NO | DAMAGE_AMOUNT |
+-----------+-----------+-----------+---------------+
| D001      | KA01M1001 |         1 |          9000 |
| D002      | KA02B2002 |        12 |         15000 |
| D003      | KA03M3003 |        12 |         10000 |
| D004      | KA04H4004 |         3 |         67800 |
| D005      | KA05A5005 |        15 |         20000 |
+-----------+-----------+-----------+---------------+
5 rows in set (0.00 sec)
```

```
mysql> INSERT INTO ACCIDENT VALUES (20, '2008-11-25', 'INDIRANAGAR');
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM ACCIDENT;
+-----------+------------+-------------+
| REPORT_NO | DDATE      | LOCATION    |
+-----------+------------+-------------+
|         1 | 2002-01-01 | RJNGR       |
|         2 | 2003-02-03 | JNGR        |
|         3 | 2003-04-05 | PEENYA      |
|        12 | 2008-05-15 | MGROAD      |
|        15 | 2008-08-20 | JPNGR       |
|        20 | 2008-11-25 | INDIRANAGAR |
+-----------+------------+-------------+
6 rows in set (0.00 sec)
```

```
mysql> -- (iv) Total Number of People Who Owned Cars Involved in Accidents in 2008
mysql> SELECT COUNT(DISTINCT O.DRIVER_ID) AS TOTAL_PEOPLE
    -> FROM OWNS O, PARTICIPATED P, ACCIDENT A
    -> WHERE O.REG_NO = P.REG_NO
    ->   AND P.REPORT_NO = A.REPORT_NO
    ->   AND YEAR(A.DDATE) = 2008;
+--------------+
| TOTAL_PEOPLE |
+--------------+
|            3 |
+--------------+
1 row in set (0.01 sec)
```

```
mysql> -- (v) Number of Accidents Involving Cars of a Specific Model (e.g., 'MARUTI')
mysql> SELECT COUNT(DISTINCT P.REPORT_NO) AS ACCIDENT_COUNT
    -> FROM PARTICIPATED P, CAR C
    -> WHERE P.REG_NO = C.REG_NO
    ->   AND C.MODEL = 'MARUTI';
+----------------+
| ACCIDENT_COUNT |
+----------------+
|              1 |
+----------------+
1 row in set (0.00 sec)
```

```
mysql>  -- (vi) Suitable Reports
mysql>   SELECT P.NAME, C.MODEL, C.REG_NO, C.YEAR
    -> FROM PERSON P, CAR C, OWNS O
    -> WHERE P.DRIVER_ID = O.DRIVER_ID
    ->   AND O.REG_NO = C.REG_NO;
+----------+---------+-----------+------+
| NAME     | MODEL   | REG_NO    | YEAR |
+----------+---------+-----------+------+
| ADARSH   | SKODA   | KA01M1001 | 2002 |
| ANOOP    | BMW     | KA02B2002 | 2008 |
| KIRAN    | MARUTI  | KA03M3003 | 2008 |
| ABHISHEK | HYUNDAI | KA04H4004 | 2001 |
| SANJAY   | AUDI    | KA05A5005 | 2008 |
+----------+---------+-----------+------+
5 rows in set (0.00 sec)
```

```
mysql>   -- 7 Complete accident history with driver name, car model, location, and damage amount
mysql>
mysql>   SELECT P.NAME, C.MODEL, A.REPORT_NO, A.DDATE, A.LOCATION, PA.DAMAGE_AMOUNT
    -> FROM PERSON P, CAR C, ACCIDENT A, PARTICIPATED PA
    -> WHERE P.DRIVER_ID = PA.DRIVER_ID
    ->   AND C.REG_NO = PA.REG_NO
    ->   AND A.REPORT_NO = PA.REPORT_NO;
+----------+---------+-----------+------------+----------+---------------+
| NAME     | MODEL   | REPORT_NO | DDATE      | LOCATION | DAMAGE_AMOUNT |
+----------+---------+-----------+------------+----------+---------------+
| ADARSH   | SKODA   |         1 | 2002-01-01 | RJNGR    |          9000 |
| ANOOP    | BMW     |        12 | 2008-05-15 | MGROAD   |         25000 |
| KIRAN    | MARUTI  |        12 | 2008-05-15 | MGROAD   |         10000 |
| ABHISHEK | HYUNDAI |         3 | 2003-04-05 | PEENYA   |         67800 |
| SANJAY   | AUDI    |        15 | 2008-08-20 | JPNGR    |         20000 |
+----------+---------+-----------+------------+----------+---------------+
5 rows in set (0.00 sec)
```

Filename: DBMS_LAB(2505260)
Directory: C:\Users\kukhr\OneDrive\Documents
Template: C:\Users\kukhr\AppData\Roaming\Microsoft\Templates\Normal.dotm
Title:
Subject:
Author: PRETSO LOMI
Keywords:
Comments:
Creation Date: 11/9/2026 9:51:00 PM
Change Number: 7
Last Saved On: 11/9/2026 10:33:00 PM
Last Saved By: PRETSO LOMI
Total Editing Time: 36 Minutes
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